READING SENSOR DATA WITH ESP8266: Introduction to Sensors for IOT

Ts. Dr. Razif bin Razali

Type of Sensors

Humidity sensors	Temperature sensors	Proximity sensors
Pressure sensors	Gyroscope sensors	Optical sensors
Chemical sensors	Gas sensors	Level sensors
Water quality sensors	Accelerometers	Air quality sensors
Image sensors	Accelerometer	Smoke sensors
Light sensors	Motion sensors	Embedded sensor modules
Flow and gas sensors	Gyroscope	Sound sensors
Environmental sensors	Infrared sensors	Biomedical sensors

Feature	Analog Sensor	Digital Sensor
Output Signal	Continuous electrical signal (e.g., voltage or current) that is proportional to the measured physical quantity.	Discrete electrical signal, typically a binary output (0s and 1s) representing the measured quantity. Often involves an internal Analog-to-Digital Converter (ADC).
Resolution	Theoretically infinite, as the output is a continuous range of values. Practical resolution can be limited by noise and the connected ADC.	Finite, determined by the number of bits used by the internal ADC. More bits mean higher resolution.
Susceptibility to Noise	More susceptible to electrical noise and interference, which can degrade the signal quality over distance.	Less susceptible to noise during transmission because the signal is already in a digital format.
Signal Processing	Often requires external signal conditioning (amplification, filtering) and an ADC to be used by digital systems (like microcontrollers).	May perform signal processing internally. Output is directly usable by digital systems.
Accuracy	Can be very accurate, but accuracy can be affected by noise and the quality of the ADC.	Accuracy is determined by the sensor's design and the internal ADC. Less prone to degradation from external factors during transmission.
Complexity	Simpler in basic construction.	Can be more complex due to internal ADC and processing capabilities.

READING SENSOR DATA WITH ESP8266: DHT11 (Temperature & Humidity Sensor)

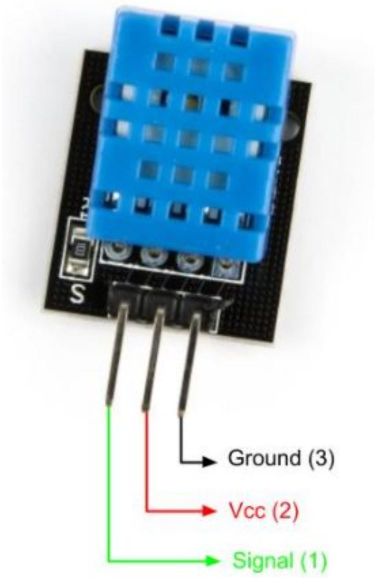
Ts. Dr. Razif bin Razali

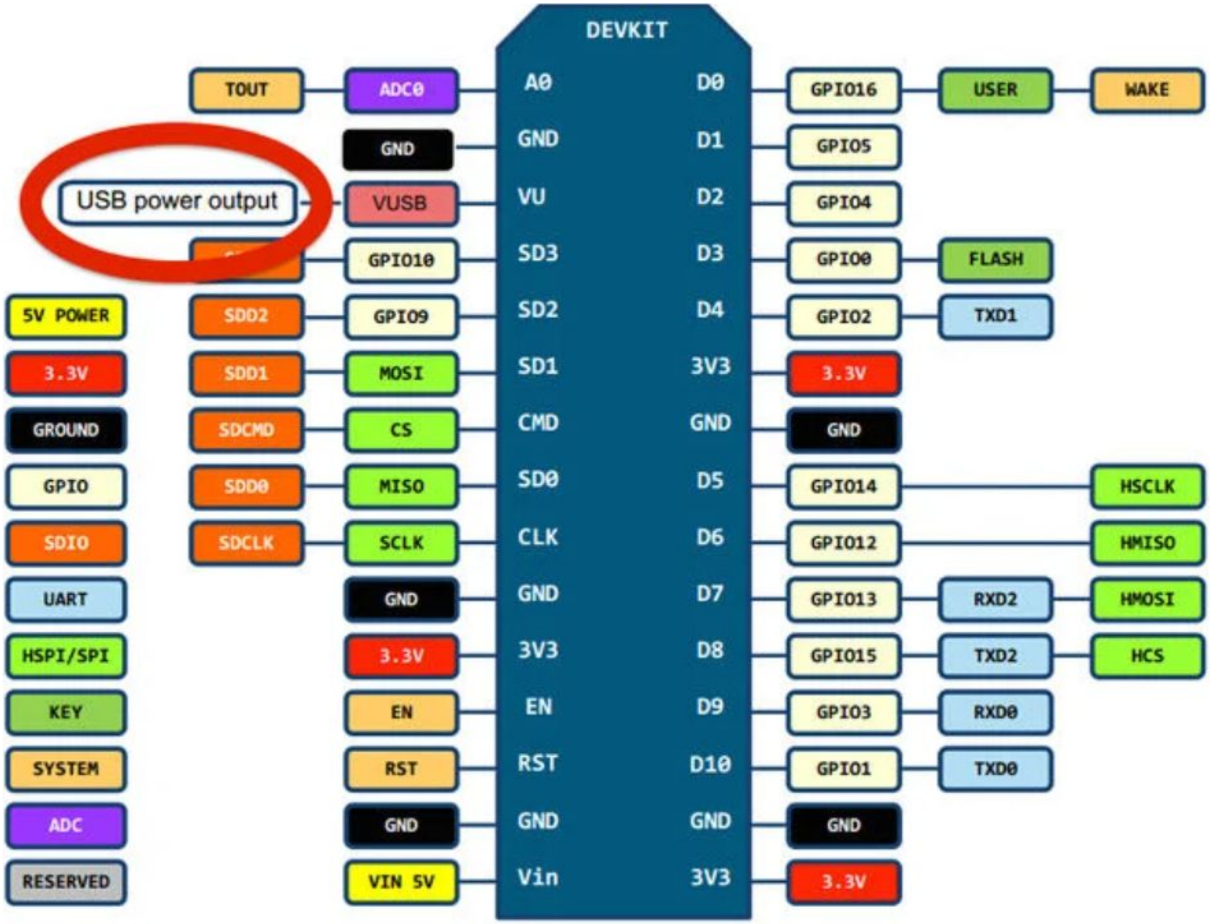
Understanding DHT11 sensor.

Ref :
https://my.cytron.io/p-dht11-sensor-module-breakout?r=1#:~:text=DHT11%20is%20the%20most%20common,with%20any%20microcontrollers%20like%20Arduino.

Datasheet :
https://drive.google.com/file/d/1Qt8ZIe-3znNOQ3vhksWffhzSIMsHC2m8/view?usp=sharing

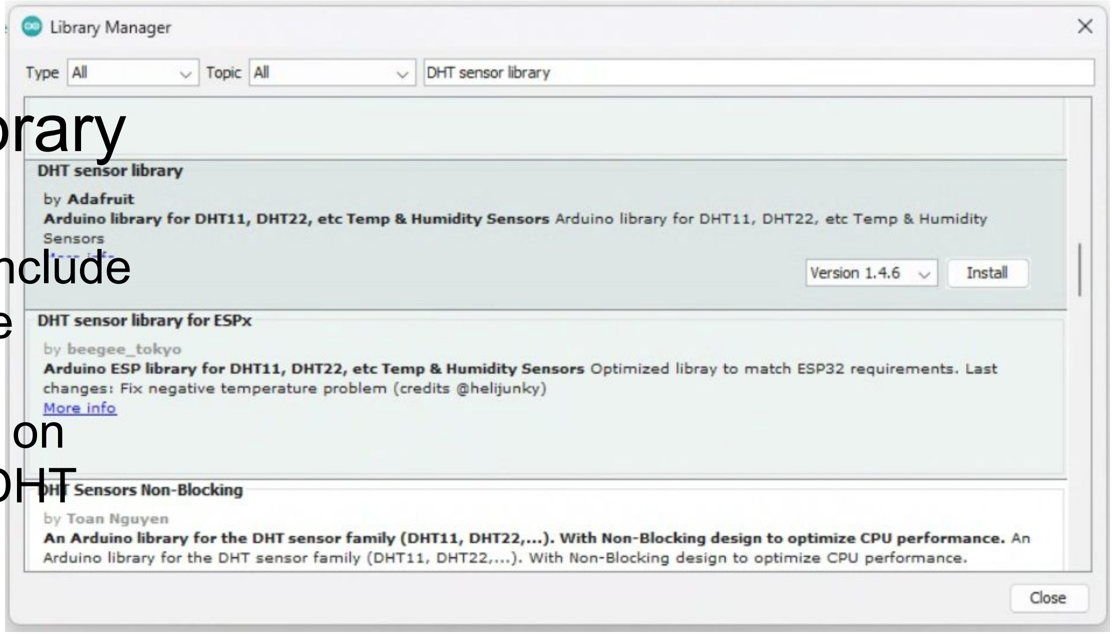
Understanding DHT11 sensor.



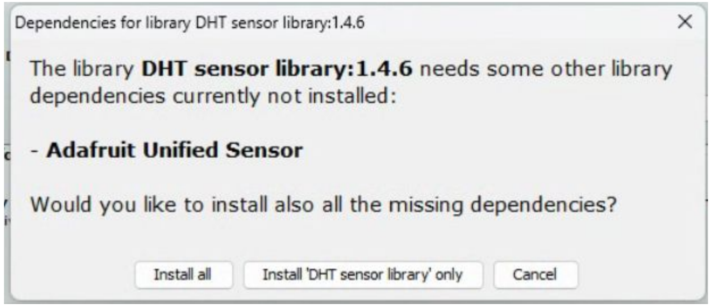


Installing DHT sensor library

- In Arduino IDE, goto Sketch > Include Library > Manage Libraries. The Library Manager should open.
- Search for “DHT sensor library” on the Search box and install the DHT library from Adafruit

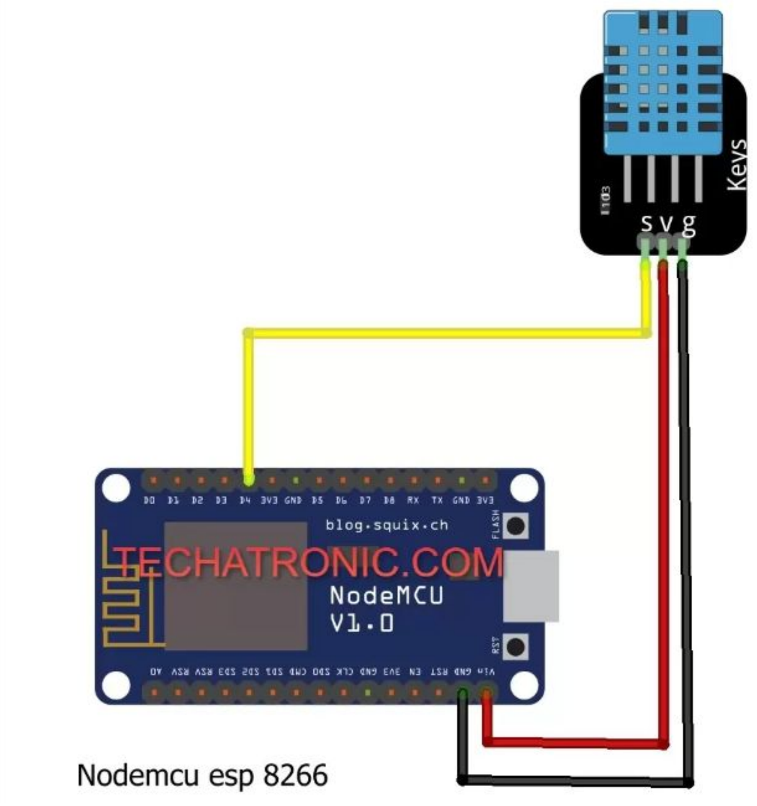


[Reference]



Display sensor readings on the Serial Monitor

- Connect the DHT11 to the ESP8266
- Copy the [github]() code, and upload to your ESP8266.
- Run, test, and debug your code and circuit.



[Reference]()

Extra Works

Read Analog Sensor (Potentiometer / LDR) data using analogRead().